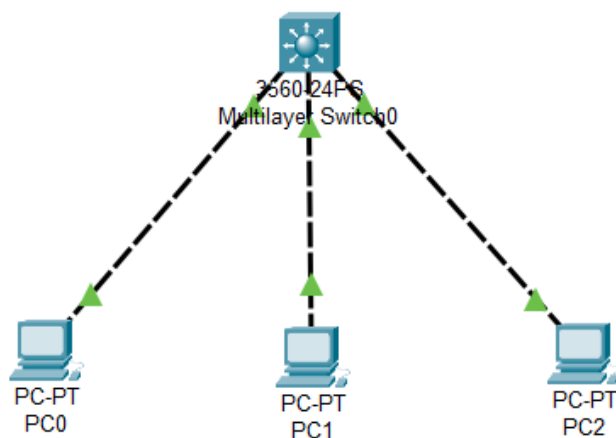


Practice 5 – Work with L3 Switch and Router

Part 1: L3 Switch

1.1 Routing between VLANs



In this scenario, we have three computers connected directly to a L3 switch. We'll segment the network into three VLANs and enable communication between them. The L3 switch used in this example is a Cisco 3560, a common L3 switch.

1.1.1 Create VLANs

Create VLANs 2, 3, and 4 by entering global configuration mode, specifying a VLAN number, and assigning a name to each VLAN.

Step 1: Create vlan with a specific number:

```
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with
CNTL/Z.
Switch(config)#vlan 2
```

Step 2: Assign a name VLAN2:

```
Switch(config-vlan)#name VLAN2
```

Step 3: Exit vlan configuration mode:

```
Switch(config-vlan)#exit
```

Step 4: Follow the same process to create vlan 3 and vlan 4, assigning them the names VLAN3 and VLAN4 respectively.

Step 5: Save configurations.

1.1.2 Assign Ports to VLANs

Similarly, for Layer 2 switches, we need to assign ports to specific VLANs based on user connections. In this case, assign port **Fa0/1** to **VLAN 2**, port **Fa0/2** to **VLAN 3**, and port **Fa0/3** to **VLAN 4**.

Step 1: Access the interface configuration mode for **FastEthernet0/1**:

```
Switch(config)#int fa0/1
```

Step 2: Configure the port as an access port to ensure it belongs to a single **VLAN**:

```
Switch(config-if)#switchport mode access
```

Step 3: Specify that FastEthernet0/1 should be a part of **VLAN 2**:

```
Switch(config-if)#switchport access vlan 2
```

Step 4: Exit interface configuration mode:

```
Switch(config-if)#exit
```

Step 5: To assign **Fa0/2** to **VLAN 3** and **Fa0/3** to **VLAN 4**, repeat the process with the respective VLAN numbers.

Step 6: Save configurations.

1.1.3 Assign IP Addresses to VLAN Interfaces

To enable inter-VLAN communication or manage the switch remotely, you need to assign IP addresses to VLAN interfaces. This is done using **SVI (Switched Virtual Interface)**.

Step 1: Select the VLAN interface:

```
Switch(config)#interface vlan 2  
Switch(config-if)#  
%LINK-5-CHANGED: Interface Vlan2, changed state to up  
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan2,  
changed state to up
```

Step 2: Assign IP to VLAN 2:

```
Switch(config-if) #ip address 2.2.2.1 255.255.255.0
```

Step 3: Exit interface configuration mode:

```
Switch(config-if) #exit
```

Step 4: Repeat the previous steps for vlan 3 and vlan 4, ensuring you assign the correct IP addresses:

1. Select **VLAN 3** interface and assign an IP address:

```
Switch(config) #interface vlan 3  
Switch(config-if) #  
%LINK-5-CHANGED: Interface Vlan3, changed state to up  
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan3,  
changed state to up  
Switch(config-if) #ip address 3.3.3.1 255.255.255.0  
Switch(config-if) #exit
```

2. Select **VLAN 4** interface and assign an IP address:

```
Switch(config) #interface vlan 4  
Switch(config-if) #  
%LINK-5-CHANGED: Interface Vlan4, changed state to up  
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan4,  
changed state to up  
Switch(config-if) #ip address 4.4.4.1 255.255.255.0  
Switch(config-if) #exit
```

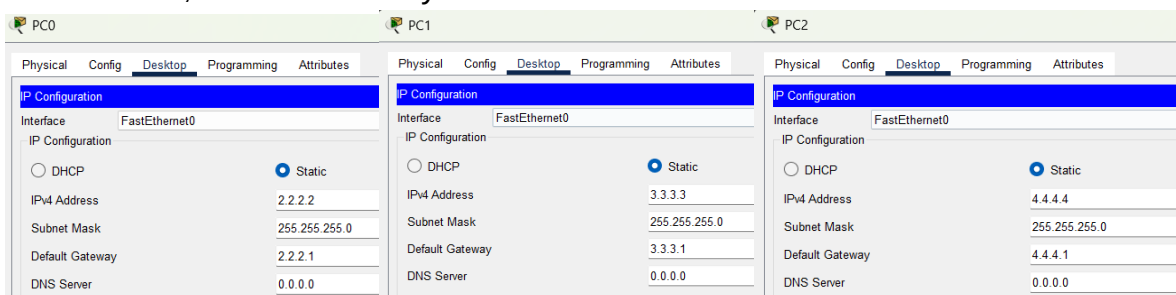
1.1.4 Configure PCs.

Assign IP addresses, subnet masks, and gateway addresses (the L3 switch's VLAN interface IP) to each computer.

PC0: 2.2.2.2, Default Gateway: 2.2.2.1

PC1: 3.3.3.3, Default Gateway: 3.3.3.1

PC2: 4.4.4.4, Default Gateway: 4.4.4.1



1.1.5 Enable IP Routing

A process that enables the L3 switch to route traffic between VLANs.

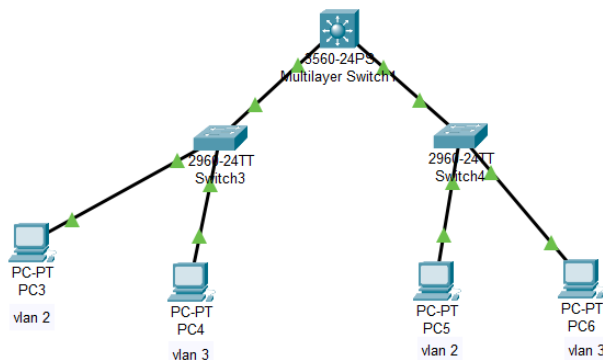
```
Switch(config)#ip routing
```

Save configurations!

Questions

1. What problem does VLAN segmentation solve in a network?
2. What does the **ip routing** command do in a L3 switch?
3. What is the difference between L2 and L3 Switches?

1.2 L3 Switch as a Distribution Layer



In this scenario, we have two L2 switches connected to a L3 switch. The L3 switch acts as a distribution layer, routing traffic between VLAN on different L2 switches.

Connect the devices using **copper straight-through cables**:

- **PC-3 → Fa0/1 on L2 Switch 3**
- **PC-4 → Fa0/2 on L2 Switch 3**
- **PC-5 → Fa0/1 on L2 Switch 4**
- **PC-6 → Fa0/2 on L2 Switch 4**
- **L2 Switch 3 → Gi0/1 on L3 Switch**
- **L2 Switch 4 → Gi0/2 on L3 Switch**

L2 Switch 3

Step 1: Assign ports to VLANs:

- Assign FastEthernet 0/1 to vlan 2:

```
Switch(config)#interface fastethernet 0/1
```

- Configure the port as an access port:
`Switch(config-if) #switchport mode access`
- Assigns the port to vlan 2:
`Switch(config-if) #switchport access vlan 2`
`% Access VLAN does not exist. Creating vlan 2`
(It's okay!)
- exit interface configuration mode:
`Switch(config-if) #exit`

Step 3: Repeat **step 2** for FastEthernet 0/2 to vlan 3:

```
Switch(config) #interface fastethernet 0/2
Switch(config-if) #switchport mode access
Switch(config-if) #switchport access vlan 3
% Access VLAN does not exist. Creating vlan 3
Switch(config-if) #exit
```

Step 4: Configure the trunk port to L3 switch:

- Select GigabitEthernet 0/1 for configuration:
`Switch(config) #interface g0/1`
- Configures the port as a trunk (allowing multiple VLANs):
`Switch(config-if) #switchport mode trunk`
- Restricts the trunk to only VLAN 2 and VLAN 3 traffic:
`Switch(config-if) #switchport trunk allowed vlan 2,3`

Step 5: Save configurations:

L2 Switch 4

Repeat all configurations from L2 Switch 3.

L3 Switch

Step 1: Configuring Trunk Ports:

- Configuring GigabitEthernet 0/1 as a Trunk Port:
`Switch>en`
`Switch#conf t`
`Switch(config) #interface gigabitEthernet0/1`

- **switchport trunk encapsulation dot1q** by this command Set **802.1Q** as the VLAN tagging protocol. **802.1Q** is the industry standard for VLAN tagging, ensuring compatibility between different switch vendors.

```
Switch(config-if) #switchport trunk encapsulation dot1q
Switch(config-if) #switchport mode trunk
```

- Restricts the trunk to allow only **VLAN 2 and VLAN 3** traffic:

```
Switch(config-if) #switchport trunk allowed vlan 2,3
```

- Repeat configuration to **gigabitEthernet 0/2**.

Step 2: Create vlan 2, 3:

- **vlan 2:**

```
Switch(config) #vlan 2
Switch(config-vlan) #name VLAN2
Switch(config-vlan) #exit
```

- **vlan 3:**

```
Switch(config) #vlan 3
Switch(config-vlan) #name VLAN3
Switch(config-vlan) #exit
```

Step3: Assign IP Addresses to VLAN Interfaces:

- Assign an IP Address to VLAN 2:

```
Switch(config) #int vlan 2
Switch(config-if) #ip address 2.2.2.1 255.255.255.0
Switch(config-if) #exit
```

- Assign IP Address to VLAN 3:

```
Switch(config) #int vlan 3
Switch(config-if) #ip address 3.3.3.1 255.255.255.0
Switch(config-if) #exit
```

Step 3: By default, Layer 3 switches don't route VLAN traffic. We must enable **ip routing**:

```
Switch(config) #ip routing
```

Step 4: Save configuration.

Step 5: Configuring PCs:

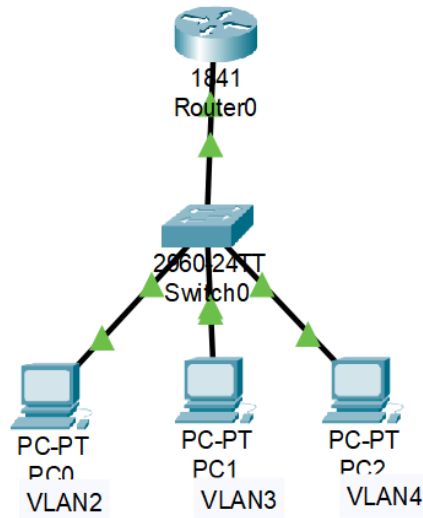
1. Open **PC-3**, go to **Desktop > IP Configuration**, and set:
 - IP Address: **2.2.2.2**

- Subnet Mask: **255.255.255.0**
- Default Gateway: **2.2.2.1**
- 2. Open **PC-5**, set:
 - IP Address: **2.2.2.3**
 - Subnet Mask: **255.255.255.0**
 - Default Gateway: **2.2.2.1**
- 3. Open **PC-4**, set:
 - IP Address: **3.3.3.2**
 - Subnet Mask: **255.255.255.0**
 - Default Gateway: **3.3.3.1**
- 4. Open **PC-6**, set:
 - IP Address: **3.3.3.3**
 - Subnet Mask: **255.255.255.0**
 - Default Gateway: **3.3.3.1**

Step 6: Testing Connectivity:

1. **Test intra-VLAN communication:**
 - Open **PC-3** and ping **PC-5** (ping 2.2.2.3).
 - Open **PC-4** and ping **PC-6** (ping 3.3.3.3).
 - If successful, it confirms that devices in the same VLAN can communicate.
2. **Test inter-VLAN communication:**
 - Open **PC-3** and ping **PC-4** (ping 3.3.3.2).
 - Open **PC-5** and ping **PC-6** (ping 3.3.3.3).
 - If successful, it confirms that inter-VLAN routing is working.

Part 2: Setting Up a Router for VLAN Communication



Switch Configuration

Step 1: Create VLANs

```
Switch>enable
Switch#conf t
Switch(config)#vlan 2
Switch(config-vlan)#name VLAN2
Switch(config-vlan)#exit

Switch(config)#vlan 3
Switch(config-vlan)#name VLAN3
Switch(config-vlan)#exit

Switch(config)#vlan 4
Switch(config-vlan)#name VLAN4
Switch(config-vlan)#end
```

- The **vlan** command creates VLANs 2, 3, and 4.
- The **name** command assigns a name to each VLAN.
- The **exit** command is used to return to the previous mode.

Step 2: Assign VLANs to Ports


```
Switch#conf t  
Switch(config)#int fa0/1  
Switch(config-if)#switchport mode access  
Switch(config-if)#switchport access vlan 2  
Switch(config-if)#exit
```

```
Switch(config)#int fa0/2  
Switch(config-if)#switchport mode access  
Switch(config-if)#switchport access vlan 3  
Switch(config-if)#exit
```

```
Switch(config)#int fa0/3  
Switch(config-if)#switchport mode access  
Switch(config-if)#switchport access vlan 4  
Switch(config-if)#end
```

Step 3: Configure Trunk Port

```
Switch#conf t  
Switch(config)#int fa0/4  
Switch(config-if)#switchport mode trunk
```

- The **switchport mode trunk** command sets the interface as a trunk.

```
Switch(config-if)#switchport trunk allowed vlan 2,3,4
```

Step 4: Save Configuration.

Router Configuration for Inter-VLAN Routing

Step 1: Enable the Router's Physical Interface

```
Router(config)#int fa0/0  
Router(config-if)#no shutdown  
Router(config-if)#exit
```

Step 2: Configure Subinterfaces for VLANs

VLAN 2 (192.168.2.0/24)

```
Router(config)#int fa0/0.2  
Router(config-subif)#encapsulation dot1Q 2
```

```
Router(config-subif)#ip address 192.168.2.1 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#exit
```

- The `int fa0/0.2` command creates a subinterface for VLAN 2.
- The `encapsulation dot1Q 2` command enables 802.1Q tagging for VLAN 2.
- The `ip address` command assigns an IP address to the subinterface.
- The `no shutdown` command enables the subinterface.

VLAN 3 (192.168.3.0/24)

```
Router(config)#int fa0/0.3
Router(config-subif)#encapsulation dot1Q 3
Router(config-subif)#ip address 192.168.3.1 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#exit
```

- Similar steps are followed for VLAN 3.

VLAN 4 (192.168.4.0/24)

```
Router(config)#int fa0/0.4
Router(config-subif)#encapsulation dot1Q 4
Router(config-subif)#ip address 192.168.4.1 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#end
```

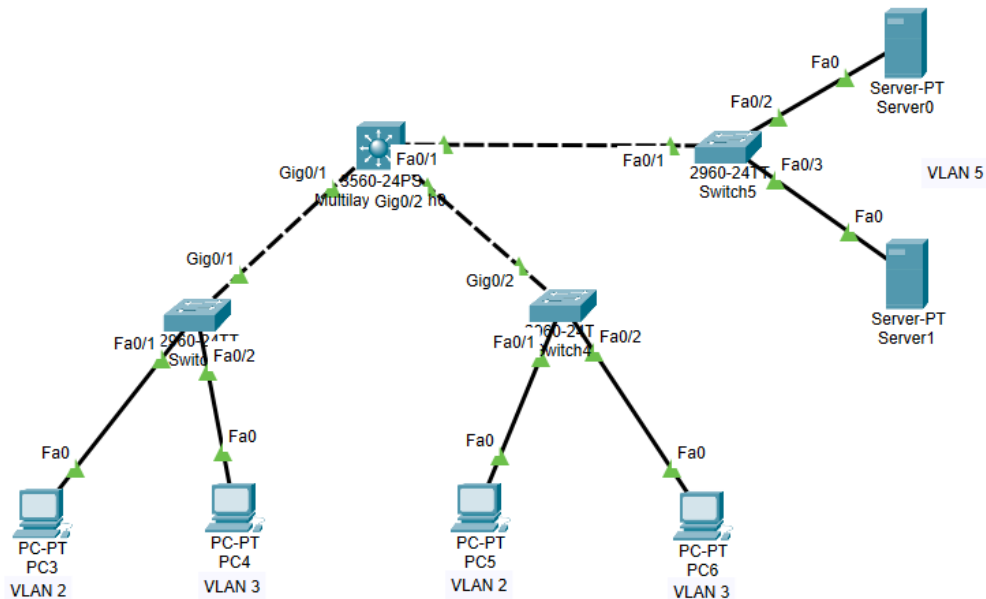
- The same configuration applies to VLAN 4.

Questions:

1. Why do we configure trunk encapsulation before enabling trunk mode on a Layer 3 switch?
2. Why does the switch automatically create a VLAN when assigning it to a port if it does not exist?
3. If ip routing is not enabled on a Layer 3 switch, how will that impact network communication?
4. What is the difference between an access port and a trunk port on a switch?
5. What is the purpose of the `encapsulation dot1Q X` command on a router?
6. Why do we need subinterfaces on a router for inter-VLAN routing?
7. What is the function of `no shutdown` command?

Task:

Modify the existing topology by adding **VLAN 5 (Server VLAN)** and ensure **full communication** between all devices in the network, including PCs and servers.



Steps to Complete:

1. **Create VLAN 5 on the L3 Switch**
 - Configure **VLAN 5** and assign it the name **ServerVLAN**.
 - Assign VLAN 5 the **IP address 5.5.5.1/24**.
2. **Configure the Switch Connected to the Servers**
 - Assign ports **Fa0/2** and **Fa0/3** to **VLAN 5**.
 - Ensure that VLAN 5 traffic is allowed on trunk links between switches.
3. **Configure IP Addresses on the Servers**
 - **Server0** → 5.5.5.2 / 255.255.255.0
 - **Server1** → 5.5.5.3 / 255.255.255.0
 - **Default Gateway** for servers: 5.5.5.1.
4. **Update Routing Settings on the L3 Switch**
 - Enable ip routing if it is not already activated.
 - Ensure that traffic is properly routed between all VLANs (VLAN 2, 3, and 5).
5. **Verify Connectivity**
 - Ensure all devices can **ping each other**, including servers (ping 5.5.5.2 and ping 5.5.5.3).